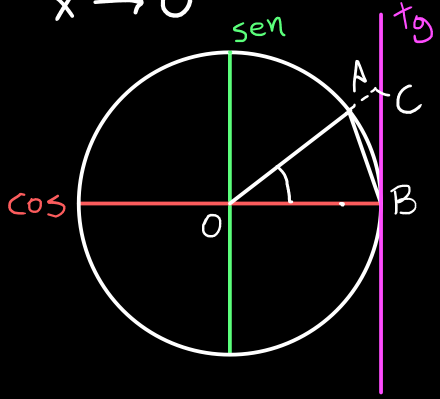
CÁLCULO

22/03

LIMITES FUNDAMENTAIS

I

$$\lim_{x \to 0} \left(\frac{\operatorname{sen} x}{x} \right) = 1$$



DEMONSTRAÇÃO

$$\text{Área } \Delta OAB = \frac{\operatorname{sen} x}{2}$$

$$\text{Área setor } \widehat{AB} = \frac{x}{2} \rightarrow \frac{\operatorname{sen} x}{\cancel{2}} < \frac{x}{\cancel{2}} < \frac{\operatorname{tg} x}{\cancel{2}}$$

$$\text{Área } \Delta OCB = \frac{\operatorname{tg} x}{2}$$

$$\operatorname{sen} x < x < \frac{\operatorname{sen} x}{\cos x}$$

$$1 < \frac{x}{\operatorname{sen} x} < \frac{1}{\cos x}$$

$$\lim_{x \to 0} 1 > \lim_{x \to 0} \frac{\operatorname{sen} x}{x} > \lim_{x \to 0} \cos x \leftarrow 1 > \frac{\operatorname{sen} x}{x} > \cos x$$

$$1 > \lim_{x \to 0} \frac{\operatorname{sen} x}{x} > 1 \rightarrow \text{TEOREMA DO CONFRONTO}$$

$$\lim_{x \to 0} \frac{\operatorname{sen} x}{x} = 1$$

II

$$\lim_{x \to 0} \left(1 + \frac{1}{x} \right)^x = e$$

DEMONSTRAÇÃO

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} \cdot a^{n-k} \cdot b^k \quad \text{ou} \quad \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} \cdot a^{n-k} \cdot b^k$$

$$\left(1 + \frac{1}{n} \right)^n = 1 + 1 + \frac{n(n-1)}{n^2} \cdot \frac{1}{2!} + \frac{n(n-1)(n-2)}{n^3} \cdot \frac{1}{3!} + \dots + \frac{n!}{n^n n!}$$

Quando $n \to \infty$:

$$\lim_{n \to \infty} \left(1 + \frac{1}{n} \right)^n = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots + \frac{1}{n!} = e \rightarrow \lim_{n \to \infty} \left(1 + \frac{1}{n} \right)^n = e$$

III $\lim_{x\to 0}\left(\frac{a^x-1}{x}\right)=\ln a$

DEMONSTRAÇÃO

$t=a^x-1 \therefore a^x=t+1$

$\ln a^x=\ln(t+1)$

$x\ln a=\ln(t+1)$

$x=\frac{\ln(t+1)}{\ln(a)}$

$\lim_{t\to 0}\frac{t}{\frac{\ln(t+1)}{\ln(a)}}$

$\ln a\cdot\lim\frac{1}{\frac{\ln(t+1)}{t}}$

$\ln a\cdot\frac{\lim_{t\to 0}1}{\lim_{t\to 0}\ln(t+1)^{\frac{1}{t}}}\Rightarrow\ln a\cdot\frac{1}{\ln\lim(t+1)^{\frac{1}{t}}}$

2º limite fundamental

$\ln a\cdot\frac{1}{\ln e}\to\ln a$

IV $\lim_{x\to 0}\frac{\cos x-1}{x}=0$

$\frac{\cos x-1}{x}\cdot\frac{\cos x+1}{\cos x+1}\to\frac{\cos^2 x-1}{x(1+\cos x)}\to\frac{\operatorname{sen}^2 x}{x(\cos x+1)}$

$\to\frac{\operatorname{sen} x}{x}\cdot\frac{\operatorname{sen} x}{\cos x+1}$

$\lim=1$

$\operatorname{sen} 0=0$

$\lim=0$

$\lim\frac{\cos x-1}{x}=0$

V $\lim_{x\to\infty}x\operatorname{sen}\left(\frac{1}{x}\right)=1$ $\quad t=\frac{1}{x}\Rightarrow x\to\infty\Rightarrow t\to 0$

$\lim_{t\to 0}\frac{\operatorname{sen}(t)}{t}=1$

$\lim\ x\cdot\operatorname{sen}\left(\frac{1}{x}\right)\cdot\frac{\frac{1}{x}}{\frac{1}{x}}\to\frac{\operatorname{sen}\left(\frac{1}{x}\right)}{\frac{1}{x}}$